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Thursday, June 26, 2025

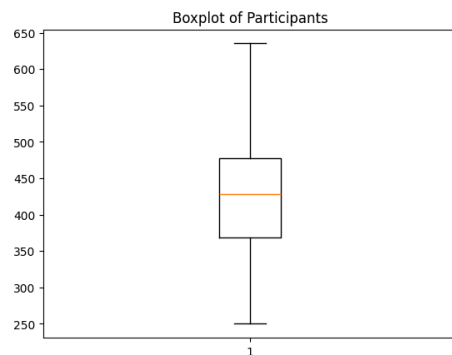
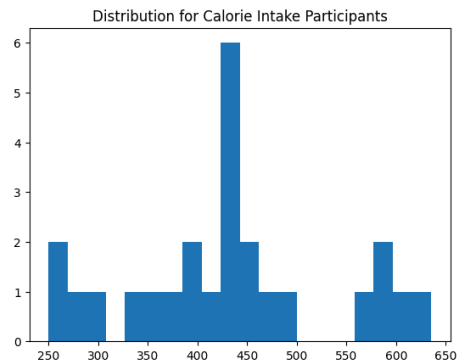
C S 329E Assignment 3

Summary of Results

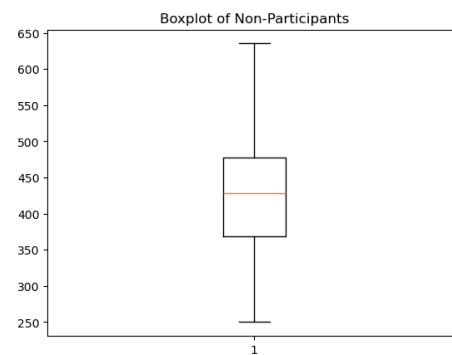
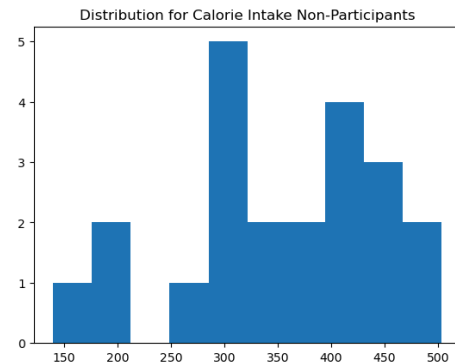
Task 1:

	Participants	Non-Participants
Count	25	22
Mean	431.40	346.80
Standard Deviation	105.70	99.50
Min	249.86	139.69
25%	368.51	290.40
50%	428.74	361.02
75%	477.96	422.57
Max	635.21	503.46

Participants:



Non-Participants:



The distributions look similar, both distributed (relatively) about their respective means (431.40, 346.80). The histogram for the non-participants appears to be skewed to the left, with more of the sampled calorie intakes appearing to the right of the mean.

Task 2:

Step 1:

| $H_0: \mu = 425$ (the mean calorie intake of participants is 425 calories)

| $H_1: \mu \neq 425$ (The mean calorie intake of participants differs from 425 calories)

| $\alpha = 0.05$

Step 2:

$n = 25$

$n < 30$, therefore the CLT does not guarantee a normal distribution

n is small and the std of the population is unknown

A t -stat, computed using the one-sample, two-sided t -test, is suitable for this hypothesis test:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Step 3:

Decision Rule: Reject H_0 if $p < \alpha$ (0.05)

Otherwise, do not reject H_0

Step 4:

$\bar{x} = 431.40$

$\mu = 425$

$s = 105.70$

$n = 25$

t -stat: 0.30

p -value: 0.76

Step 5:

There is no sufficient evidence to reject H_0 as p is greater than 0.05.

We cannot conclude that μ differs from 425.

Task 3:

The 90% confidence interval for the mean calorie intake for participants is (395.23, 467.57).

This means that 90% of sampled means ($\bar{x}$) from the population will fall between 395.23 and 467.57 calories.

Task 4:

Step 1:

μ_p : participants

μ_{np} : non participants

| $H_0: \mu_p = \mu_{np}$ (participants consumed the same number of calories as non-participants)

| $H_1: \mu_p > \mu_{np}$ (participants consumed more calories than non-participants)

| $\alpha = 0.05$

Step 2:

$n_1 = 25, n_2 = 22$

n is small and the std of the population is unknown

A t -stat, computed using the two-sample, one-sided t -test, is suitable for this hypothesis test:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\mu_p - \mu_{np} = d$, d is set to 0, therefore the formula can be written as such:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step 3:

Decision Rule: Reject H_0 if $p < \alpha$ (0.05)

Otherwise, do not reject H_0

Step 4:

$n_1 = 25, n_2 = 22$

$$\bar{x}_1 = 431.40, \bar{x}_2 = 346.80$$
$$s_1 = 105.70, s_2 = 99.50$$

t -stat: 2.82

p -value: 0.0035

Step 5:

Reject H_0 as p is less than 0.05 in favor of the alternative hypothesis.

We have significant evidence at the $\alpha = 0.05$ level that the participants consumed more calories than non-participants.

Task 5:

Independence: the two sets of children seem to have no influence on each other, but there is no evidence for or against a true random sample of the two distinct populations.

Same measurement: there is no evidence for or against the same measurement method being used, however, given that this is a small experiment with participants and non-participants, we would assume that the researchers would not modify their data collection methods across populations.

Approximately Normal Distribution: neither sample includes any outliers, and while the non-participant histogram appears slightly skewed, the distributions are similar.

Python Code

(see following page)

Task 1

Summarize the data by whether children participated in the meal preparation or not. Use an appropriately labelled table to show the results. Also include a graphical presentation that shows the distribution of calories for participants vs. non-participants. Describe the shape of each distribution and comment on the similarity (or lack thereof) between the distributions in each population.

```
[3]: # getting the data
participants = data['Calorie intake for participants']
non_participants = data['Calorie intake for non-participants']
non_participants = non_participants.dropna()
```

```
[4]: # Calorie intake participants
print(participants.describe())
print()
print(non_participants.describe())
```

count 25.000000 ***

```
[5]: # participants
plt.hist(participants)
plt.title('Distribution for Calorie Intake Participants')
plt.show()

fig1, ax1 = plt.subplots()
ax1.set_title('Boxplot of Participants')
ax1.boxplot(participants)
plt.show()
```

<Figure size 640x480 with 1 Axes> ***

```
[6]: # non-participants
plt.hist(non_participants)
plt.title('Distribution for Calorie Intake Non-Participants')
plt.show()

fig2, ax2 = plt.subplots()
ax2.set_title('Boxplot of Non-Participants')
ax2.boxplot(non_participants)
plt.show()
```

<Figure size 640x480 with 1 Axes> ***

Task 2

Does the mean calorie consumption for those who participated in the meal preparation differ from 425? Formally test at the $\alpha=0.05$ level using the 5 steps outlined in the module

```
[7]: # try out a 2-sided hypothesis t-test
# H_0:  $\mu = 425$ 
# H_1:  $\mu \neq 425$ 

results = stats.ttest_1samp(participants, 425)

print("T-statistic: ", results.statistic)
print("P-value: ", results.pvalue)
```

T-statistic: 0.30272115216243584 ***

Task 3

Calculate a 90% confidence interval for the mean calorie intake for participants in the meal preparation. Interpret the confidence interval.

```
[8]: # perform a two-sided hypothesis t-test and retrieve
# the resulting confidence interval

result = stats.ttest_1samp(participants, np.mean(participants), alternative='two-sided')
interval = result.confidence_interval(confidence_level = 0.90)

print(f"Confidence Interval: ({interval[0]:.2f}, {interval[1]:.2f})")
```

Confidence Interval: (395.23, 467.57) ***

Task 4

Formally test whether or not participants consumed more calories than non-participants at the $\alpha=0.05$ level using the 5 steps outlined in our class lecture notes

```
[9]: # try out a two-sample one-sided t-test
#  $\mu_p$ : participants
#  $\mu_{np}$ : non-participants

# H_0:  $\mu_p = \mu_{np}$ 
# H_1:  $\mu_p > \mu_{np}$ 

statistic, pvalue = stats.ttest_ind(participants, non_participants, equal_var=False, alternative="greater")
print("T-statistic: ", statistic)
print("P-value: ", pvalue)
```

T-statistic: 2.8248363392676477 ***